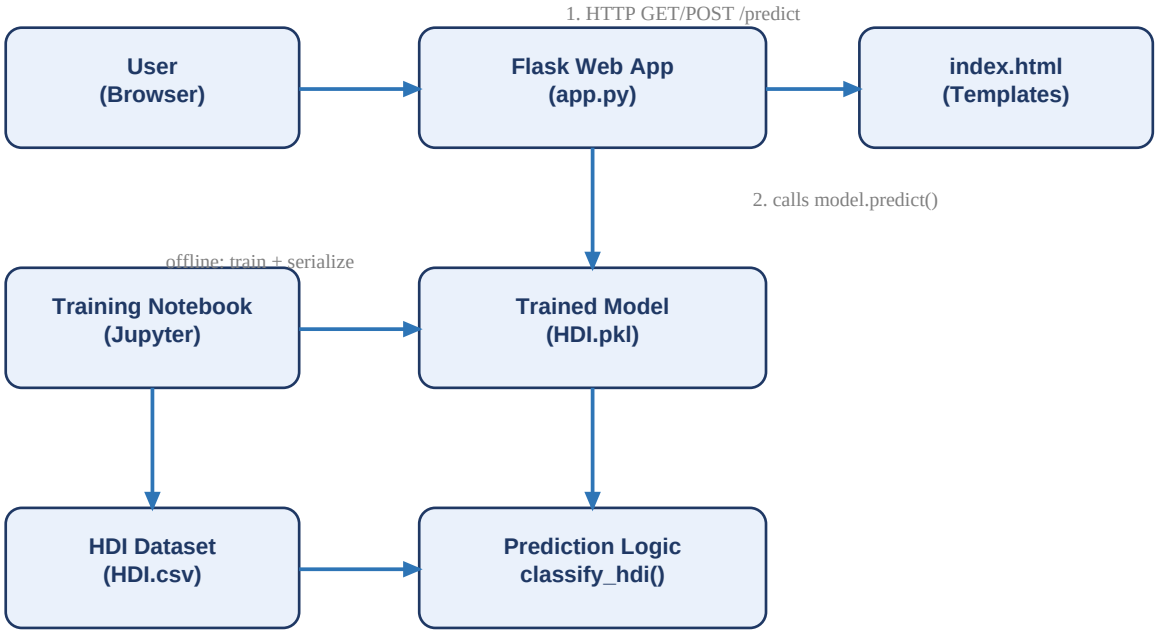
System Design Diagram

HDI Prediction System | Project Design Phase

High-Level Architecture

The system is split into an offline training path and an online serving path. The training notebook reads the historical UNDP dataset (HDI.csv), fits a regression model that maps the four core indicators to the HDI value, and serializes it as HDI.pkl. At runtime, the Flask application (app.py) loads this file once, renders the input form (index.html), and computes predictions on demand without touching the raw dataset again.

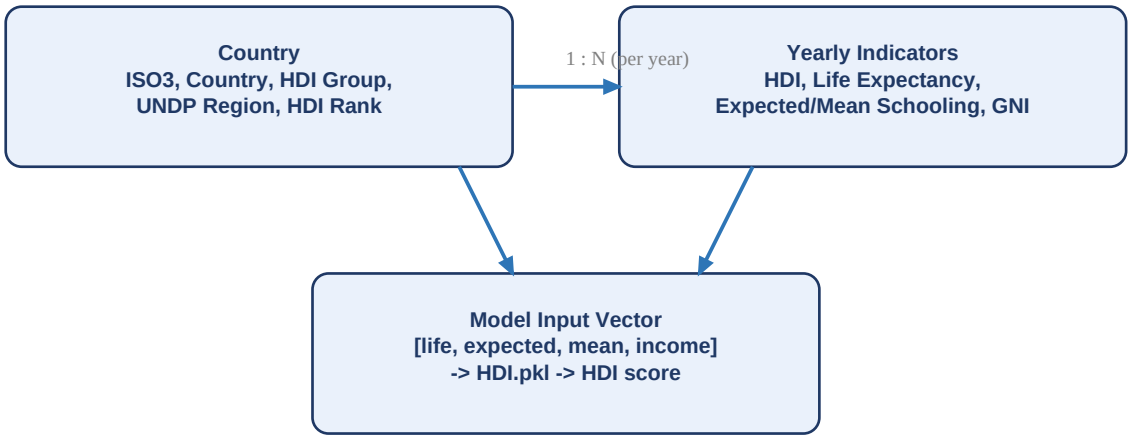


Request Flow

- User opens the web page; Flask's **/** route renders index.html.
- User submits life expectancy, expected years of schooling, mean years of schooling, and GNI per capita through the form (or a JSON POST to **/predict**).
- app.py builds a NumPy feature vector and calls **model.predict()** on the loaded HDI.pkl model.
- The raw prediction is clipped to the valid HDI range [0, 1] and passed through **classify_hdi()** to obtain the UNDP tier label (Very High / High / Medium / Low).
- The result is returned either as a rendered HTML page or as JSON, depending on the request type.

Entity / Data Relationship Diagram (HDI Dataset)

The HDI.csv dataset is conceptually organized around a **Country** entity with one row per country, and a wide set of **Yearly Indicator** columns (one column per indicator per year, 1990–2021). At inference time, only the most recent values of four indicators are reduced into a single input vector consumed by the trained model.



Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript (Flask templates + static assets)
Backend / API	Python, Flask
ML Model	scikit-learn regression model, serialized with pickle (HDI.pkl)
Data Processing	pandas, numpy (used in the training notebook)
Deployment	Gunicorn WSGI server (per requirements.txt)